Mission 7: The Case File

The story

"Day 90. Arben's family doctor in Fier asks for his file. Six missions have each saved a piece of his story: the ambulance, the road, the hospital, the clinic. Your last mission puts them together. One command, one patient, the whole journey, from the minute he was last seen well to his last test."

What you need to know

Nothing new about medicine. One idea about time:

- Mission 4 saved **planned** times, as if the hospital were empty.
- Mission 5 saved **actual** times, after waiting in line.
- **Waited** = actual − planned.

The base challenge

Your goal: one command that tells a patient's whole story, using only what the other missions saved, and never breaks when a part is missing.

Goal 1: Show a case

`case <case id>` reads the case from your database and prints it in four sections: triage, route, hospital and recovery. It uses only what the other missions saved. It does not run the route or the hospital simulation again.

Example CLI interaction (you may design yours differently):

```
synapse> case P-0001
=== Case P-0001: Arben Hoxha, 67, Fier ===

TRIAGE
07:30  Last seen well
08:10  Case created by the ambulance crew
       Symptoms: cannot-speak, cannot-understand, right-arm-weak, right-face-droop
       Severity 8 (moderate) · likely blocked pipe: left-middle (big)
       Drug possible until 12:00 · clot removal possible until 07:30 tomorrow

ROUTE
08:10  Leaves Fier: Fier -> Durres -> Tirana, 110 min
10:00  Arrives at Qendra Neuro Tirana

HOSPITAL         PLANNED   ACTUAL   WAITED
Scan             -         10:10    -
Drug             10:30     10:40    10 min
Clot removal     11:00     11:10    10 min

RECOVERY (normal for ages 65-79)
Day 14   reaction 540 ms well below · memory 4 below      · colour words 31.0 s well below
Day 45   reaction 470 ms below      · memory 5 normal     · colour words 26.0 s below
Day 90   reaction 430 ms normal     · memory 6 normal     · colour words 22.0 s normal
```

Goal 2: Handle a case that is not complete

Not every patient goes through every mission. Print what exists and say plainly what is missing. Never crash on a missing part.

```
synapse> case P-0003
=== Case P-0003: Besnik Kola, 58, Kukes ===

TRIAGE
09:00  Last seen well
10:00  Case created by the ambulance crew
       Symptoms: left-arm-weak, ignores-left-side
       Severity 5 (moderate) · likely blocked pipe: unknown
       Drug possible until 13:30 · clot removal not possible

ROUTE
10:00  Leaves Kukes: Kukes -> Lezhe -> Shkoder, 140 min
12:20  Arrives at Spitali Rozafa

HOSPITAL
Not recorded: only Qendra Neuro Tirana is simulated. Planned drug: 12:50.

RECOVERY
No clinic visits yet.
```

A case with no route prints `ROUTE: not planned yet. Run route P-0005.`, and the sections after it say the same kind of thing.

Goal 3: Find a case

`case find <text>` searches every case by name and city. The search ignores upper and lower case and matches part of a word. If exactly one case matches, show it in full. If several match, list them and let the user pick one.

```
synapse> case find ar
2 cases match "ar":
1. P-0001  Arben Hoxha   67  Fier
2. P-0002  Drita Marku   72  Shkoder
Enter a number to open, or press ENTER to go back: 2
=== Case P-0002: Drita Marku, 72, Shkoder ===
...

synapse> case find KORCE
1 case matches "KORCE". Opening it.
=== Case P-0004: Mira Dedja, 61, Korce ===
...
```

Error handling

Your program must handle and clearly report errors such as:

- "ERROR: Missing case id. Usage: case <case id>"
- "ERROR: No case called "P-0099"."
- "ERROR: Search text cannot be empty."
- "No cases match "xyz"." (a message, not a crash)
- "ERROR: The database file data/cases.json is damaged. No cases loaded."



The bonus challenge

For extra points.

Bonus 1: Export for the next hospital

`case export <case id>` writes two files into an `exports/` folder, creating the folder if needed: `P-0001.txt` with the same text as `case P-0001`, and `P-0001.json` with the whole case. Running it again replaces both files.

```
synapse> case export P-0001
SUCCESS: Wrote exports/P-0001.txt and exports/P-0001.json.
```

Bonus 2: The board

`case board` prints every case on one screen, one row each, so a doctor sees the whole day at a glance. For the last visit, count the results below normal. If you built Mission 6, Bonus 2, mark a flagged case with `!`.

```
synapse> case board
ID      NAME         WENT TO              FIRST TREATMENT     WAITED        LAST VISIT
P-0001  Arben Hoxha  Qendra Neuro Tirana  drug 10:40          10 min        day 90: all normal
P-0002  Drita Marku  Qendra Neuro Tirana  clot removal 09:30  0 min         day 90: 2 below normal  !
P-0003  Besnik Kola  Spitali Rozafa       drug 12:50 planned  not recorded  no visits
P-0004  Mira Dedja   Qendra Neuro Tirana  clot removal 13:10  20 min        no visits
! = check for a new stroke
```

Bonus 3: Time is brain

`case cost <case id>` turns the patient's time into the number from the Overview: about **1.9 million brain cells** for every minute of a large stroke without treatment. Count the minutes from last seen well to the first treatment (drug or clot removal, actual if recorded, else planned), and the minutes spent waiting in the hospital line.

```
synapse> case cost P-0001
Last seen well 07:30 -> first treatment 10:40 (drug): 190 min
About 361 million brain cells lost before treatment, at 1.9 million a minute.
10 of those minutes were spent waiting in the hospital line: about 19 million.
```

The 1.9 million figure is a published average for large strokes, used here as an estimate.

What the judges will test

- `case` for every case in the examples, including Besnik's incomplete one.
- A case that stopped after triage, with no route and no visits.
- Closing and reopening the program, then printing the same cases.
- `case find` with upper case, part of a word, a city, and text that matches nothing.
- A damaged `cases.json`.

What this mission saves

Nothing new. This is the finished product: every mission's saved work in one place.



The whole journey

This is one patient's journey through the finished program, from the ambulance to the last clinic visit. The numbers are examples; each mission defines its exact rules.

```
$ python synapse.py
Synapse 1.0 · type "help" to see all commands
SUCCESS: Atlas loaded from data/atlas.txt: 16 pipes, 12 areas.
SUCCESS: Symptoms loaded from data/symptoms.txt: 14 symptoms.

synapse> triage new                                    (Mission 3)
Patient name: Arben Hoxha
Age: 67
City: Fier
Time now (HH:MM): 08:10
Last seen well (HH:MM): 07:30
Was that yesterday? (yes/no): no
Symptoms (comma separated): cannot-speak, cannot-understand, right-arm-weak, right-face-droop
SUCCESS: Case P-0001 created and saved.
  Severity:            8 (moderate)
  Likely blocked pipe: left-middle (big)
  Drug possible:       yes, until 12:00 (3 h 50 min left)
  Clot removal:        yes, until 07:30 tomorrow

synapse> lesion block left-middle                      (Mission 2)
Blocked pipe: left-middle (big)
Pipes without blood (3): left-middle, left-middle-top, left-middle-bottom
Areas that stop working (3): speaking, right-arm-face, understanding
Expected symptoms (4): cannot-speak, right-arm-weak, right-face-droop, cannot-understand

synapse> route P-0001                                  (Mission 4)
Fastest hospital that can remove the clot: Qendra Neuro Tirana
  Route:         Fier -> Durres -> Tirana, 110 min
  Arrives:       10:00
  Drug:          10:30
  Clot removal:  11:00
SUCCESS: Route saved for P-0001.

synapse> unit run data/today.events                    (Mission 5)
08:30  P-0002 arrives
...
10:00  P-0001 arrives
10:10  P-0001 scan starts
10:40  P-0001 drug given (40 min after arrival)
11:10  P-0001 clot removal starts
...
SUCCESS: Hospital times saved for P-0001, P-0002, P-0004.

synapse> clinic visit P-0001                           (Mission 6)
Days since the stroke: 90
Test 1 of 3: reaction time ...      Average: 430 ms
Test 2 of 3: memory span ...        Longest row repeated: 6 digits
Test 3 of 3: colour words ...       22.0 s
SUCCESS: Visit on day 90 saved for P-0001.

synapse> case P-0001                                   (Mission 7)
=== Case P-0001: Arben Hoxha, 67, Fier ===
```




```
TRIAGE
07:30  Last seen well
08:10  Case created: severity 8 (moderate), likely blocked pipe left-middle (big)

ROUTE
08:10  Leaves Fier: Fier -> Durres -> Tirana, 110 min
10:00  Arrives at Qendra Neuro Tirana

HOSPITAL         PLANNED   ACTUAL   WAITED
Scan             -         10:10    -
Drug             10:30     10:40    10 min
Clot removal     11:00     11:10    10 min

RECOVERY (normal for ages 65-79)
Day 14   reaction 540 ms well below · memory 4 below      · colour words 31.0 s well below
Day 45   reaction 470 ms below      · memory 5 normal     · colour words 26.0 s below
Day 90   reaction 430 ms normal     · memory 6 normal     · colour words 22.0 s normal
```

The labels in brackets show which mission builds each command; they are not part of the output.